rapidmesh

A Hierarchical Bottom-Up Centroidal-Voronoi Meshing Algorithm
for B-Rep Solids with Analytic and NURBS Geometry

Specification and mathematical derivation

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Abstract

This document specifies the meshing algorithm rapidmesh uses to turn a boundary representation (B-Rep) of one or more material regions into a conforming, isotropic, high-quality tetrahedral mesh. The algorithm is *dimensionally hierarchical*: it meshes 0-cells (corners), then 1-cells (edges), then 2-cells (faces), then the 3-cell (volume), and *freezes* each level before the next consumes it. Within every dimension it combines *error-driven adaptive sampling* (insert where a geometric tolerance is violated) with *sizing-weighted Centroidal Voronoi Tessellation* (Lloyd relaxation) for point quality. The frozen surface triangulation is a *hard constraint* on the volume Delaunay, which is what makes the boundary watertight by construction. We derive, with a computer-algebra system, the arc-length, curvature, deflection-sizing, first/second fundamental form and distance-faithful chart formulae for every edge and face modality the geometry kernel supports (line, circle/arc, surface–surface intersection, NURBS curve, extruded profile; plane, cylinder, cone, sphere, torus, extruded/ruled, NURBS), and give pseudocode for each stage.

Contents

1 Design philosophy

Principle 1 (Dimensional hierarchy)**.** A k-cell is meshed only after every $(k-1)$-cell on its boundary is meshed and *frozen*. Corners pin edges; edges pin faces; faces pin the volume. Shared lower-dimensional entities are meshed once and reused, so adjacent cells agree on their common boundary exactly (conformity across seams is inherited, not reconstructed).

Principle 2 (Two orthogonal mechanisms per level)**.** Each dimension uses (i) *error-driven adaptivity* to decide *how many* points and *where* (insert at the worst geometric-error location, then re-relax, until a tolerance is met), and (ii) *CVT/Lloyd relaxation* to decide *point quality* (move each free site to the density-weighted centroid of its (restricted) Voronoi cell). Adaptivity controls density; CVT controls regularity.

Principle 3 (Constrained volume, not recovered boundary)**.** The frozen surface triangulation enters the volume stage as a set of *constraint facets*. The volume tetrahedralization is required to contain them as faces. The boundary is therefore *enforced*, not statistically recovered: there are no straddling tetrahedra, hence no "noses" protruding into a void and no gaps. This is the decisive structural property and the reason the surface must be completed *before* the volume is seeded.

The geometry source is a B-Rep assembled from the exact CSG arrangement: faces are trimmed analytic (or NURBS) surfaces, edges are analytic curves (including the intersection curves booleans create), vertices are corner points, and the radial-edge topology records which faces meet along each edge and the material region on each side. The mesh is built *from this geometry*, independent of any input tessellation.

2 Notation and preliminaries

2.1 The sizing field

A scalar *sizing field* $h : \mathbb{R}^3 \to \mathbb{R}_{>0}$ prescribes the target element edge length. It is assembled bottom-up (edges set it on 1-cells, faces refine it on 2-cells, the volume smooths it through the interior) and is always made *Lipschitz* (gradation-limited):

$$h(\mathbf{x}) \;\leq\; h(\mathbf{y}) + \alpha\,\|\mathbf{x}-\mathbf{y}\| \qquad \forall\,\mathbf{x},\mathbf{y}, \tag{1}$$

with grading constant $\alpha \in (0,1]$ (the default $\alpha = \frac{1}{2}$ grows neighbouring elements by at most $\sim 1.5\times$). Equation (1) is the standard mesh-gradation constraint [16]; it is enforced by a fast-marching / sweep over the size sources.

2.2 Centroidal Voronoi Tessellation

Given a density $\rho : \Omega \to \mathbb{R}_{>0}$ on a domain Ω and sites $\{\mathbf{z}_i\}_{i=1}^n$, the Voronoi region of $\mathbf{z}_i$ is $V_i = \{\mathbf{x} \in \Omega : \|\mathbf{x} - \mathbf{z}_i\| \le \|\mathbf{x} - \mathbf{z}_j\| \ \forall j\}$. The tessellation is *centroidal* when each site coincides with the mass centroid of its region,

$$\mathbf{z}_i \ = \ \mathbf{c}_i \ := \ \frac{\displaystyle\int_{V_i} \rho(\mathbf{x})\,\mathbf{x}\,\mathrm{d}\mathbf{x}}{\displaystyle\int_{V_i} \rho(\mathbf{x})\,\mathrm{d}\mathbf{x}}. \tag{2}$$

A CVT is a critical point of the energy $\mathcal{F}(\{\mathbf{z}_i\}) = \sum_i \int_{V_i} \rho(\mathbf{x})\,\|\mathbf{x} - \mathbf{z}_i\|^2\,\mathrm{d}\mathbf{x}$ [1]. *Lloyd's algorithm* [2] alternates "recompute V_i" and "move $\mathbf{z}_i \leftarrow \mathbf{c}_i$" and converges to a CVT; the resulting point set is blue-noise / isotropic, which is precisely the input a Delaunay mesher needs for well-shaped elements.

Density that realises a sizing field. To make the local point spacing follow h, choose

$$\rho(\mathbf{x}) \ = \ h(\mathbf{x})^{-(d+2)} \ \ \text{(theory, } d \text{ dimensions)}, \qquad \rho(\mathbf{x}) \ = \ h(\mathbf{x})^{-d} \ \text{(used in practice)}, \tag{3}$$

because a CVT equidistributes the energy density, giving local spacing $\propto \rho^{-1/(d+2)}$; the simpler h^{-d} (number density $\propto$ points per unit volume) is robust and is what we weight cells by. See [3, 4] for the exponent discussion.

2.3 Discrete centroids and the restricted setting

On a triangulation, the integral (2) is approximated by summing over the simplices incident to $\mathbf{z}_i$, each weighted by its measure times the density at its centroid. In dimension d with incident simplices T of centroid $\mathbf{g}_T$, measure $|T|$,

$$\mathbf{c}_i \ \approx \ \frac{\sum_{T \ni \mathbf{z}_i} |T| \ \rho(\mathbf{g}_T)\,\mathbf{g}_T}{\sum_{T \ni \mathbf{z}_i} |T| \ \rho(\mathbf{g}_T)}. \tag{4}$$

On a curved surface, the relevant object is the *restricted* Voronoi diagram (Voronoi cells clipped to the surface) and its dual *restricted Delaunay* triangulation [6, 7]; in practice we relax in a 2D chart (§5) so (4) applies in the chart and the move is lifted back onto the surface.

2.4 Robust predicates

All combinatorial decisions (orientation, in-circle/in-sphere) use exact adaptive-precision predicates [14]; only non-decision quantities (centroid weights, sizing values) are floating point. Map/iteration order is made deterministic (seedless hashing), so the mesh is reproducible.

3 Stage 0 – Vertices

Every B-Rep corner $\mathbf{p}$ becomes a *pinned* site: it never moves and is present in every downstream triangulation. Pinning corners preserves sharp features exactly. Formally these are the 0-cells of the cell complex and the fixed points of all relaxations below.

4 Stage 1 – Edges

A B-Rep edge is a curve $\mathbf{C} : [t_0, t_1] \to \mathbb{R}^3$ between two corners. We support the modalities in Table 1.

Modality	$\mathbf{C}(t)$	Source
Line	$\mathbf{p}_0 + t\,\mathbf{d}$	straight B-Rep edge
Circle / arc	$\mathbf{c} + r(\cos t\,\mathbf{x} + \sin t\,\mathbf{y})$	rim, fillet, boolean of quadrics
Intersection	implicit $\mathbf{S}_a \cap \mathbf{S}_b$, seeded by the chain	generic surface–surface boolean
NURBS	$\dfrac{\sum_i N_{i,p}(t)\,w_i\,\mathbf{P}_i}{\sum_i N_{i,p}(t)\,w_i}$	imported / profile curve [17]
Extruded profile	$\mathbf{b} + z\,\mathbf{a} + u_x\,P_x(t) + u_y\,P_y(t)$	lifted 2D profile (airfoil)
Polyline	piecewise linear chain	faceted fallback

Table 1: Edge curve modalities. All expose $\mathbf{C}(t), \mathbf{C}'(t), \mathbf{C}''(t)$.

4.1 Arc length

The arc length and the natural (unit-speed) parameter are

$$s(t) = \int_{t_0}^{t} \left\|\mathbf{C}'(\tau)\right\|\, \mathrm{d}\tau, \qquad \mathrm{d}s = \left\|\mathbf{C}'(t)\right\|\, \mathrm{d}t. \tag{5}$$

We tabulate $s(t)$ once (adaptive Gauss/Simpson) and invert $t(s)$ by table lookup + a Newton step using $\mathrm{d}t/\,\mathrm{d}s = 1/\left\|\mathbf{C}'\right\|$, so points can be placed at prescribed arc-length spacings. For a circle $\|\mathbf{C}'\| = r$ and $s = r\,\Delta t$ exactly; for a line it is exact; for NURBS / extruded profiles the table is used.

4.2 Curvature

For a space curve the (unsigned) curvature is

$$\kappa(t) \;=\; \frac{\|\mathbf{C}'(t) \times \mathbf{C}''(t)\|}{\|\mathbf{C}'(t)\|^3}, \qquad R(t) = \frac{1}{\kappa(t)} \text{ (radius of curvature)}, \tag{6}$$

derived from the Frenet apparatus and verified symbolically. A straight segment has $\kappa \equiv 0$, $R \equiv \infty$; a circle of radius r has $\kappa \equiv 1/r$.

4.3 Deflection sizing

The chord between two samples a parametric distance apart deviates from the true arc by the *sagitta*. For an arc of radius R subtending a half-angle $\theta/2$, with chord length $\ell = 2R\sin(\theta/2)$, the maximum chord–arc deviation is

$$\varepsilon \;=\; R\big(1 - \cos\tfrac{\theta}{2}\big) \;=\; \frac{R\,\theta^2}{8} + O(\theta^4) \;\approx\; \frac{\ell^2}{8R}. \tag{7}$$

Solving the small-angle relation for the admissible edge length given a *deflection tolerance* ε gives the local target

$$\boxed{\ell(\mathbf{x}) \;=\; \sqrt{8\,\varepsilon\,R(\mathbf{x})}} \qquad \Big(\text{exact: } \ell = 2\sqrt{R^2 - (R-\varepsilon)^2}\Big), \tag{8}$$

both verified with sympy. The edge sizing field is then

$$h_{\mathrm{edge}}(\mathbf{x}) \;=\; \min\!\Big(h_{\max},\; \sqrt{8\,\varepsilon\,R(\mathbf{x})}\Big), \tag{9}$$

i.e. refine where the curve bends (R small), coast at $h_{\max}$ where it is straight. This is the curve analogue of the surface bound (13).

4.4 1D CVT and error-driven adaptivity

On a fixed edge the n interior sites $s_1 < \cdots < s_n$ (arc-length coordinates, endpoints pinned) relax by the 1D centroid rule (the segment analogue of (4)): with neighbours s_{i-1}, s_{i+1} and density $\rho = h^{-1}$,

$$s_i \;\leftarrow\; \frac{\rho_{i-\frac{1}{2}}\,\frac{s_{i-1}+s_i}{2} + \rho_{i+\frac{1}{2}}\,\frac{s_i+s_{i+1}}{2}}{\rho_{i-\frac{1}{2}} + \rho_{i+\frac{1}{2}}}, \qquad \rho_{i\pm\frac{1}{2}} = h\big(\tfrac{s_i+s_{i\pm1}}{2}\big)^{-1}, \tag{10}$$

which equidistributes spacing $\propto h$. Adaptivity wraps the relaxation:

Algorithm 1 MESHEDGE($\mathbf{C}, h, \varepsilon$)

sample $s(t)$ table; $L \leftarrow s(t_1)$
$n \leftarrow \max\big(1,\ \lceil \int_0^L h(s)^{-1}\,\mathrm{d}s \rceil\big)$ ▷ initial count from the sizing field
place $s_1, \ldots, s_n$ equidistributing h (cumulative-ρ inversion)
repeat
  for a few iterations **do** relax all s_i by (10)
  end for
  $e^\star$, $s^\star \leftarrow \max_i$ deflection of the chord $[\mathbf{C}(s_i), \mathbf{C}(s_{i+1})]$ vs. the arc, and its location
  if $e^\star > \varepsilon$ **then** insert a site at $s^\star$; $n \mathrel{+}= 1$
  end if
until $e^\star \le \varepsilon$ **and** max site move $< \tau_{\text{move}}$
return $\{\mathbf{C}(s_i)\}$ ▷ frozen edge points; corners included

The relax–insert loop is the 1D instance of the general scheme: relax for quality, insert at the worst error, repeat until the tolerance holds. Shared edges are meshed once and cached, so both incident faces read identical points.

5 Stage 2 – Faces

A face is a trimmed surface $\mathbf{S} : \Omega \subset \mathbb{R}^2 \to \mathbb{R}^3$, $(u, v) \mapsto \mathbf{S}(u, v)$, bounded by oriented loops of frozen edges. The strategy is: parameterise the face by a *distance-faithful chart*, relax interior points there with a 2D surface-CVT (boundary = frozen edge points, held fixed), insert where the surface deflection tolerance is violated, and lift every chart point exactly onto $\mathbf{S}$. The output is a *conforming triangulation of the face* that the volume stage will treat as a hard constraint.

5.1 First fundamental form and area

With $\mathbf{S}_u = \partial\mathbf{S}/\partial u$, $\mathbf{S}_v = \partial\mathbf{S}/\partial v$, the metric (first fundamental form) coefficients and area element are

$$E = \mathbf{S}_u\cdot\mathbf{S}_u, \quad F = \mathbf{S}_u\cdot\mathbf{S}_v, \quad G = \mathbf{S}_v\cdot\mathbf{S}_v, \qquad \mathrm{d}A = \sqrt{EG - F^2}\;\mathrm{d}u\,\mathrm{d}v. \tag{11}$$

A curve $t \mapsto (u(t), v(t))$ in parameter space has length $\int \sqrt{E\dot{u}^2 + 2F\dot{u}\dot{v} + G\dot{v}^2}\;\mathrm{d}t$; this is the metric the chart must reproduce for Euclidean 2D operations (separation, centroid) to mean arc length on the surface. Table 2 lists E, F, G, $\mathrm{d}A$ for every analytic modality, all derived with sympy.

5.2 Curvature and surface deflection sizing

The shape operator's eigenvalues are the principal curvatures κ_1, κ_2; from the second fundamental form $L = \mathbf{S}_{uu}\cdot\mathbf{n}$, $M = \mathbf{S}_{uv}\cdot\mathbf{n}$, $N = \mathbf{S}_{vv}\cdot\mathbf{n}$ ($\mathbf{n} = \mathbf{S}_u \times \mathbf{S}_v / \|\mathbf{S}_u \times \mathbf{S}_v\|$), the Gaussian and mean curvatures are

$$K = \kappa_1\kappa_2 = \frac{LN - M^2}{EG - F^2}, \qquad H = \tfrac{1}{2}(\kappa_1 + \kappa_2) = \frac{EN - 2FM + GL}{2(EG - F^2)}, \tag{12}$$

Surface	$\mathbf{S}(u,v)$	E	F	G	$\mathrm{d}A$
Plane	$\mathbf{o}+u\mathbf{e}_1+v\mathbf{e}_2$	1	0	1	$\mathrm{d}u\,\mathrm{d}v$
Cylinder (θ,h)	$r(\cos\theta,\sin\theta)+h\mathbf{a}$	r^2	0	1	$r\ \mathrm{d}u\,\mathrm{d}v$
Sphere (λ,φ)	$r(\cos\varphi\cos\lambda,\cos\varphi\sin\lambda,\sin\varphi)$	$r^2\cos^2\varphi$	0	r^2	$r^2\,\lvert\cos\varphi\rvert\ \mathrm{d}u\,\mathrm{d}v$
Cone (θ,s)	$s\tan\alpha(\cos\theta,\sin\theta)+s\mathbf{a}$	$s^2\tan^2\alpha$	0	$\sec^2\alpha$	$\lvert s\rvert\tan\alpha\sec\alpha\ \mathrm{d}u\,\mathrm{d}v$
Torus (θ,ϕ)	$(R{+}r\cos\phi)(\cos\theta,\sin\theta){+}r\sin\phi\,\mathbf{a}$	$(R{+}r\cos\phi)^2$	0	r^2	$r\,\lvert R{+}r\cos\phi\rvert\ \mathrm{d}u\,\mathrm{d}v$
Extruded (t,h)	$\mathbf{b}+P(t)+h\mathbf{a}$	$\lVert P'(t)\rVert^2$	0	1	$\lVert P'(t)\rVert\ \mathrm{d}u\,\mathrm{d}v$
NURBS (u,v)	tensor B-spline [17]	$\mathbf{S}_u{\cdot}\mathbf{S}_u$	$\mathbf{S}_u{\cdot}\mathbf{S}_v$	$\mathbf{S}_v{\cdot}\mathbf{S}_v$	$\sqrt{EG{-}F^2}$

Table 2: First fundamental form per modality ($\mathbf{a}$ the unit axis; α the cone half-angle). All but the NURBS row are diagonal ($F=0$), which is what makes the charts below clean.

and $\kappa_{1,2}=H\pm\sqrt{H^2-K}$. Verified values (Table 3) confirm the geometric intuition: developables have $K=0$, the sphere is umbilic.

Surface	K	principal curvatures	$\kappa_{\max}=1/R_{\min}$
Plane	0	$0,\ 0$	0
Cylinder	0	$1/r,\ 0$	$1/r$
Cone	0	$1/(s\tan\alpha\sec\alpha),\ 0$	$1/(s\tan\alpha\sec\alpha)$
Sphere	$1/r^2$	$1/r,\ 1/r$	$1/r$
Torus	$\dfrac{\cos\phi}{r(R+r\cos\phi)}$	$\dfrac{1}{r},\ \dfrac{\cos\phi}{R+r\cos\phi}$	$1/r$
Extruded	0	$\kappa_P(t),\ 0$	$\kappa_P(t)=\dfrac{\lvert P'\times P''\rvert}{\lVert P'\rVert^3}$

Table 3: Principal/Gaussian curvature per modality (sympy-verified). $R_{\min}=1/\kappa_{\max}$ feeds the surface deflection bound.

The surface sagitta argument mirrors (7) with $R=R_{\min}=1/\kappa_{\max}$: a triangle edge of length ℓ on a surface of smallest principal radius $R_{\min}$ deviates from the surface by $\varepsilon\approx\ell^2/(8R_{\min})$, so the face sizing field is

$$h_{\text{face}}(\mathbf{x}) \;=\; \min\Bigl(h_{\max},\ \sqrt{8\,\varepsilon\,R_{\min}(\mathbf{x})}\Bigr),\qquad R_{\min}=1/\kappa_{\max}. \tag{13}$$

5.3 Distance-faithful charts

A chart $\Phi:$ face $\to\mathbb{R}^2$ lets the planar CVT machinery act on the surface. We want Φ as close to an *isometry* as possible so Euclidean 2D distance equals surface arc length.

Developables (cylinder, cone, extruded): exact isometry. A surface with $K\equiv 0$ unrolls into the plane without distortion.

- *Cylinder*: $\Phi(\theta,h)=(r\theta,\ h)$. Then $E=G=1, F=0$ in the chart – exact.
- *Cone*: polar unroll $\Phi(\theta,s)=\bigl(\rho\cos\psi,\rho\sin\psi\bigr)$ with slant $\rho=s/\cos\alpha$ and sweep $\psi=\theta\sin\alpha$. The induced chart metric is $E=s^2\tan^2\alpha,\ F=0,\ G=\sec^2\alpha$ – identical to the cone's first fundamental form (Table 2), i.e. *exactly isometric* (sympy-verified).
- *Extruded*: $\Phi(t,h)=(\sigma(t),\,h)$ with $\sigma(t)=\int^t\lVert P'\rVert$ the profile arc length; $E=G=1, F=0$ – exact.

Sphere: azimuthal-equidistant. The sphere ($K = 1/r^2 \neq 0$) admits no isometry to the plane. We use the azimuthal-equidistant chart about a cap centre: the chart radial coordinate is the exact geodesic distance $r\psi$ (ψ the geodesic colatitude), so radial distances are exact; the circumferential direction stretches by

$$\mathrm{stretch}(\psi) = \frac{\psi}{\sin\psi} = 1 + \frac{\psi^2}{6} + \frac{7\psi^4}{360} + O(\psi^6), \tag{14}$$

(sympy-verified) which is ≤ 1.05 for caps up to $\psi \approx 32°$ and stays bounded well past a hemisphere. The chart is a bijection on any cap that excludes the antipode; bijectivity is validated by a round-trip test $\mathrm{dist}\big(\mathrm{proj}_{\mathbf{S}}(\mathbf{p}),\, \Phi^{-1}\Phi(\mathbf{p})\big) < \mathrm{tol}$.

General NURBS / non-developable: metric-scaled or restricted CVT. For a NURBS patch we scale the parameters by the local metric ($\sqrt{E}, \sqrt{G}$) at the patch centroid (a first-order isometry) and, when a single chart distorts too much, fall back to a genuine restricted-Voronoi relaxation on the surface [5, 4]: the centroid (4) is computed from the restricted cells and the move is projected back onto $\mathbf{S}$ by closest-point.

Revolution faces and periodicity. A full revolution (cylinder/cone/torus barrel) is periodic in θ; its boundary loops are rim circles that do not close in the unrolled rectangle. Such a face is meshed by a structured (θ, v) grid whose rings reuse the *rim* longitudes (radial alignment, no seam) and whose row spacing follows h in arc length – the uniform grid is the on-surface CVT optimum of a developable. A sphere cap meeting an intersection circle is itself a revolution about the line of centres (the circle lies in a plane $\perp$ that axis), so the same latitude/longitude construction applies, with both incident caps deriving their longitude reference from the *shared* rim so their rings align.

5.4 Surface CVT and adaptivity

Algorithm 2 MESHFACE($\mathbf{S}$, loops, h, ε)

build chart Φ (Table-driven by modality); validate bijectivity
$B \leftarrow \Phi$(frozen edge points of all loops) ▷ fixed boundary in chart
scatter interior sites in Φ(face) at spacing h (greedy graded Poisson) inside the trim loops
repeat
  for a few iterations **do**
    2D Delaunay of $B \cup \{\text{interior}\}$ (exact predicates)
    move each interior site by (4), $\rho = h^{-2}$, clamp inside the loops; keep B fixed
  end for
  $e^\star \leftarrow$ max triangle deflection vs. $\mathbf{S}$ (sagitta, (13))
  if $e^\star > \varepsilon$ **then** insert a site at the worst triangle's circumcentre (in the chart)
  end if
until $e^\star \leq \varepsilon$ **and** max move $< \tau_{\mathrm{move}}$
lift: $\mathbf{p}_k = \mathbf{S}(\Phi^{-1}(\text{chart site}_k))$ – exact on surface
return conforming triangulation of the face (its triangles + vertices)

After Stage 2 every face carries a frozen, conforming triangulation; shared edges guarantee face-to-face conformity. The union of all face triangulations is a watertight surface mesh $\mathcal{S}$.

6 Stage 3 – Volume

6.1 Sizing field from the surface

The volume sizing field is grown from the completed surface: at each surface vertex the local feature size and the surface element size seed h_{vol}, which is then propagated through the interior and gradation-limited by (1). Crucially this is computed *after* the surface is final, so it already reflects curvature- and feature-driven refinement, including *local thickness*: where two boundary sheets are closer than $h_{\max}$ (a thin web, a blunt trailing edge, a slender bore wall) the field shrinks so the interior is resolved.

6.2 Interior seeding

Interior sites are thrown at the field density: a randomised, separation-guarded dart pattern (or a body-centred-cubic lattice graded by h [15]) with a per-point clearance from $\mathcal{S}$ of one local element, and a global budget $\int_\Omega h^{-3}\,\mathrm{d}V$ so a sharp feature does not inflate the count. Surface vertices are *not* re-seeded; they are the boundary of the volume point set.

6.3 3D CVT and quality-driven adaptivity

Interior sites relax by the 3D centroid rule (4) with $\rho = h^{-3}$; surface vertices stay fixed. A site whose centroid leaves the domain is mirrored back across the nearest boundary plane. After relaxation, sites are inserted where the local edge length exceeds $h_{\max}$ or where element quality (minimum dihedral angle, radius–edge ratio) is poor, and the relaxation is repeated.

6.4 Boundary-constrained tetrahedralization

This is the structural heart of the method. The volume is the *constrained Delaunay tetrahedralization* (CDT) of the interior sites *plus* the frozen surface mesh $\mathcal{S}$ as constraint facets [13, 12, 10]:

$$\mathrm{Tetrahedralize}\big(\{\text{interior sites}\} \cup V(\mathcal{S})\big) \quad \text{s.t. every } \triangle \in \mathcal{S} \text{ is a union of tet faces.} \tag{15}$$

Because $\mathcal{S}$ is watertight and every constraint triangle is recovered as tetrahedron faces, each surface triangle has a tetrahedron exactly on each side; region labels are then assigned by flood fill across non-constraint faces starting from the region tag attached to each surface triangle's two sides. No tetrahedron straddles the boundary, so:

Proposition 1 (Watertightness). *If $\mathcal{S}$ is a closed, non-self-intersecting, conforming surface mesh and the CDT recovers all of its facets, then for every constraint triangle the two incident tetrahedra carry the region labels prescribed by that triangle's front/back tags, and the boundary of each meshed region is exactly the corresponding subset of $\mathcal{S}$. In particular there are no protrusions into a void and no gaps.*

Facet recovery in a CDT may require a bounded number of Steiner points *on the constraint facets*; these are inserted on $\mathcal{S}$ (never off it), so the boundary geometry is preserved exactly [12, 13]. Sliver tetrahedra (near-zero volume, good circumradius) that survive Delaunay refinement are removed by a final quality pass – vertex smoothing constrained to carriers, plus flips/exudation [11, 9] – which is volume- and conformity-preserving because boundary vertices move only within their surface/edge carrier.

Algorithm 3 MESHVOLUME($\mathcal{S}, h$)

build h_{vol} from $\mathcal{S}$ (feature size, thickness), gradation-limit by (1)
seed interior sites at density h_{vol}^{-3}, clearance $\geq h$ from $\mathcal{S}$
repeat
    CDT of (interior $\cup\, V(\mathcal{S})$) with $\mathcal{S}$ as constraints
    for a few iterations **do** relax interior sites by (4), $\rho = h^{-3}$
    end for
    insert sites where edge length $> h_{\max}$ or quality is poor
until size and quality targets met **and** max move $< \tau_{\text{move}}$
flood-fill region labels across non-constraint faces from surface tags
quality post-pass (carrier-constrained smoothing, flips); drop slivers
return tetrahedra, per-tet region, boundary faces ($= \mathcal{S}$)

7 The sizing field, end to end

The single field h threads all stages and is only ever *refined* downward in size:

1. **Edges** set h from curve deflection (9) and $h_{\max}$ / per-region overrides.
2. **Faces** take the min of the edge field and the surface deflection bound (13); the smoothed result drives the surface CVT.
3. **Volume** takes the min of the face field and the local-thickness / feature-size estimate, then gradation-limits (1) through the interior.

Because each stage only ever lowers h and then re-smooths, the field is monotone and Lipschitz, and the element-size contract (a documented $1.5\times$ max-edge bound, as in gmsh) holds globally.

8 Why this guarantees conformity

The earlier rapidmesh path built one *unconstrained* Delaunay over all points and recovered the boundary by labelling each tetrahedron with the region of its centroid. That recovers the boundary only *statistically*: where the surface sample is coarser than a feature, a tetrahedron straddles the true surface, and whole-tet labelling then produces a protrusion (centroid inside) or a gap (centroid outside). The present algorithm removes the failure mode at its root by (a) completing a watertight, error-controlled surface mesh *first*, and (b) making that mesh a *constraint* on the volume rather than a target to be rediscovered. Conformity is then a theorem (Proposition 1), not a sampling-density gamble. The cost is the constrained Delaunay / facet recovery, which is exactly the machinery the method must keep; the variational (CVT) relaxation supplies the well-shaped interior that keeps recovery cheap and slivers rare [8, 9].

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